

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 2 – CASE STUDY

Course Code:	CSA06	Course Title:	Design and Analysis of Algorithms
CO Assessed:	CO3: Articulation (BL4, BL5)	Assessment:	Assessment Tool 2 – Case Study
Weightage:	20% of CIA	Total Marks:	100
CO3 Statement:	Design Divide and Conquer algorithms for Merge Sort, Quick Sort, Binary Search and Matrix Multiplication		
Student Name:	C Shahanaj	Reg. No.:	192372292
Section / Batch:	CSE AI/2023	Date:	5/08/2026

Instructions to Students

- Answer the following question. Total Marks: 100.
- Carefully analyze the given case study questions.
- Explain in detail with sample code if needed.
- Clearly identify all key issues, provide an in-depth analysis, and propose innovative, feasible solutions with strong justification.

Question Type	Focus Area	Questions
Case Study	Focus on Design Divide and Conquer algorithms: Merge Sort, Quick Sort, Binary Search and Matrix Multiplication	Q1

SECTION A – CASE STUDY

Code of Conduct

I _____shahanaj/192372292_____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____shahanaj_____

RUBRICS FOR CALCULATION

Case Study					
Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Case	25%	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding ; problem not identified correctly
Application of Concepts	25%	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts
Analysis	20%	In-depth analysis with	Good analysis with logical reasoning	Basic analysis with limited	Weak or no

		strong reasoning and insights		depth	analysis; lacks reasoning
Solution Design	20%	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided
Clarity	10%	Clear, well structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand

Marks Evaluation:

Criteria	Wt.	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Case	25%				
Application of Concepts	25%				
Analysis	20%				
Solution Design	20%				
Clarity	10%				
Total Marks (100):					

Faculty In-charge	Course Coordinator	Head of Department
Name:	Name:	Name:
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

CODE FOR QUICK SORT:

```
#include <stdio.h>
```

```
void quickSort(int a[], int low, int high);
```

```
int partition(int a[], int low, int high);
```

```
int main()
```

```
{
```

```
    int n, a[1000], i;
```

```
    scanf("%d",&n);
```

```
    for(i=0;i<n;i++)
```

```
        scanf("%d",&a[i]);
```

```
    quickSort(a,0,n-1);
```

```
    for(i=0;i<n;i++)
```

```
        printf("%d ",a[i]);
```

```
        return 0;
    }

    void quickSort(int a[], int low, int high)
    {
        if(low<high)
        {
            int p=partition(a,low,high);
            quickSort(a,low,p-1);
            quickSort(a,p+1,high);
        }
    }
```

```
    int partition(int a[], int low, int high)
    {
        int pivot=a[high];
        int i=low-1,j,temp;

        for(j=low;j<high;j++)
        {
            if(a[j]<=pivot)
            {
                i++;
                temp=a[i];
                a[i]=a[j];
                a[j]=temp;
            }
        }
```

```
}
```

```
temp=a[i+1];
```

```
a[i+1]=a[high];
```

```
a[high]=temp;
```

```
return i+1;
```

```
}
```

OUTPUT:

**SIMATS**
ENGINEERING

SIMATS Online Programming Lab
DAA PROGRAMMING

CHERUVU SHAHANAJ
192372292RA

Q1
Cryptocurrency Price Tracker
(Quick Sort)
100 marks

0/1 answered

A cryptocurrency platform tracks and sorts prices dynamically for thousands of coins. Analyze how Quick Sort can be applied for fast sorting. Identify issues such as pivot selection and fluctuating data. Apply divide and conquer concepts to explain its working. Analyze performance and propose optimizations with proper justification.

Your Code

```
1 #include<stdio.h>
2 void quickSort(int a[],int low,int high);
3 int partition(int a[],low,int high);
4 int main()
5 {
6     int n,a[1000];
7     scanf("%d",&n);
8     //int a[n];
9     for(i=0;i<n;i++)
10         scanf("%d",&a[i]);
11     quickSort(a,0,n-1);
12     for(i=0;i<n;i++)
13         printf("%d ",a[i]);
14     return 0;
```

Input

```
6
45 12 89 23 67 10
```

Output

```
System error.
```